



DC – AI – Routing – Switching – Security – Wireless – Automation

Construction of a multi-geo multi-cluster

An HPE Juniper Networking Techpost

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HPE



Contents

1	Multi-Geo Overlay Procedure	3
1.1	Introduction	3
1.2	Pre-Requisites	4
1.2.1	Jump host	4
1.2.2	Management Cluster	5
1.2.3	Workload Clusters	5
1.3	Preparing the Jump host	6
1.3.1	Kubeconfigs	6
1.4	Preparing the Registries (Air-gapped)	7
1.5	Installing Submariner 0.20.0	8
1.6	Installing Karmada 1.13.1	10
1.6.1	Preparing the Karmada Context	13
1.6.2	Joining the Workload Clusters	14
1.7	Operational Tips	15
1.7.1	Application Support	15
1.7.2	Karmada kubeconfig	15
1.7.3	Template File for Multi-geo	15
1.7.4	Switchover vs Failover	15
1.7.5	Monitoring and Troubleshooting	16
1.8	Conclusion	17
1.9	Useful links	17
1.10	Glossary	17
1.11	Acknowledgements	18

List of Figures

1	Logical Representation of a Kubernetes Cluster	3
2	A Multi-Geo Multi-Cluster	4

1 Multi-Geo Overlay Procedure

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The procedure for building a multi-geography multi-cluster from three Red Hat OpenShift Container Platform (RHOCP) clusters. The constructed multi-cluster will be capable of supporting Broadband Edge (BBE) cloud-native applications such as BNG CUPS Controller and Address Pool Manager (APM) in a geo-redundant capacity.

1.1 Introduction

A typical Kubernetes cluster is comprised of at least 3 control-plane sites/nodes and 3 worker sites/nodes. For cost and simplicity, worker and control-plane functions may be combined into a hybrid node at the expense of redundancy. The cluster is accessed from a remote host through the Kubernetes API and the container registry (for pushing images). The remote host is often referred to as a jump or bastion host (see Figure 1).

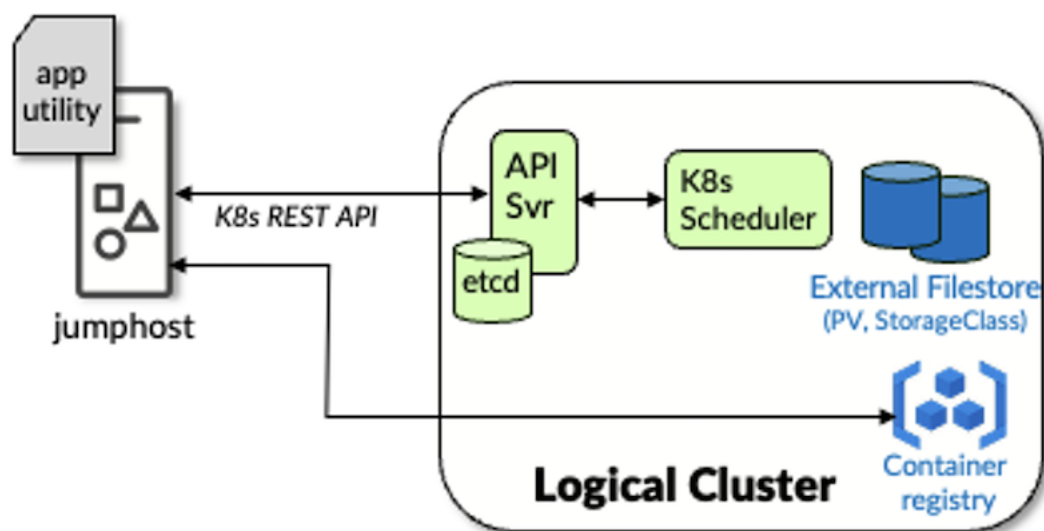


Figure 1: Logical Representation of a Kubernetes Cluster

Multi-geography redundancy for Juniper Cloud-native Network Functions (CNFs) such as APM and BNG CUPS Controller, requires a topology of three Kubernetes clusters. Each cluster has at least three control-plane sites and three worker sites (or simply 3 combined control-plane/worker or hybrid nodes) to support basic redundancy and availability requirements. Each cluster is in its own geography or availability zone.

One cluster is designated as the Management Cluster and has reachability to the other two clusters which are designated as Workload Clusters. All three clusters are joined into a multi-cluster with the addition and setup of Karmada and Submariner Open Source Software (OSS) packages. A separate jumphost or bastion host with access to all three clusters will be used as the installation platform (see Figure 2).

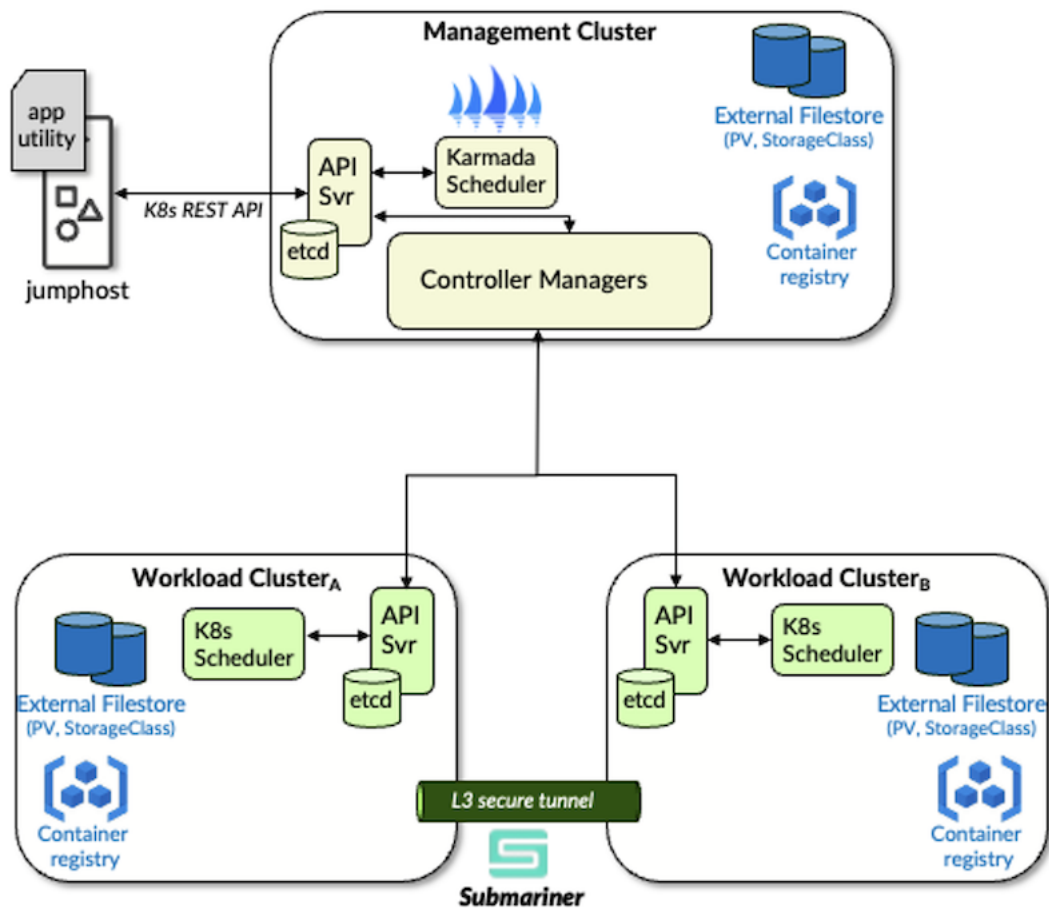


Figure 2: A Multi-Geo Multi-Cluster

The procedure outlined in this document describes the steps needed to install and deploy the OSS such that a multi-geo multi-cluster is realized. Instructions for performing air-gapped and non-air-gapped installation are provided. Procedures for air-gapped are denoted in **light-blue text**.

1.2 Pre-Requisites

1.2.1 Jumphost

A bastion host or jump host serves as a secure location for installing, configuring, and operating the OSS on the K8s clusters that will form the multi-cluster. The jumphost has administrative access to the K8s REST APIs of all three clusters.

Hardware dimensions

- CPU: 2 cores
- Memory: 8 GiB
- Storage: 128 GiB

Software dimensions

- OS: Ubuntu 22.04 LTS
- User account with sudo privileges

Network access

For non-air-gapped installation, Internet access will be needed to download OSS components. Specifically, to the following sites:

- <https://github.com/karmada-io/karmada/releases/download/>
- <https://raw.githubusercontent.com/karmada-io/karmada/master>
- <https://github.com/submariner-io/releases/releases/download>

The jumphost must be able to access the Container registry and the K8s REST API of the Management Cluster and each of the Workload Clusters.

1.2.2 Management Cluster

The Management Cluster hosts the Karmada OSS in a separate Kubernetes context. The regular context can host CNF application components. The Management Cluster is a redundant cluster (minimum of three hybrid nodes). The Dimensions of each node are as follows.

Hardware Dimensions

- CPU: 8 cores
- Memory: 24 GiB
- Storage: 256 GiB

Software Dimensions

- OS: RHOC 4.16 or later with
 - CNI: OVN
 - Registry: Openshift Image Registry
 - NLB: MetalLB
- Pod & Service CIDRs: default

Network Access

Internet access will be needed to download OSS components. Specifically, to the following sites:

- docker.io
- registry.k8s.io
- quay.io

The Management Cluster must be able to access the K8s REST API of each Workload Cluster.

1.2.3 Workload Clusters

The Workload Clusters host the bulk of the Multi-geo enabled CNF Application micro-services. Each Workload Cluster is a redundant cluster (minimum of three hybrid nodes). The Dimensions of each node are as follows:

Hardware Dimensions

- CPU: 16 cores (if only APM application deployment, this can be scaled down to 8 cores)

- Memory: 64GiB
- Storage: 512 GiB

Software Dimensions

- OS: RHOC 4.16 or later with
 - CNI: OVN
 - Registry: Openshift Image Registry
 - NLB: MetalLB
 - CSI: Longhorn
- Pod & Service CIDRs: Each Workload cluster must have different (non-overlapping) POD and Service CIDRs (since Submariner will join the two Workload Cluster internal networks, each internal network address space must not overlap with the other)

Network Access

Internet access will be needed to download OSS components. Specifically, to the following sites:

- quay.io

Each Workload Cluster must be able to reach its peer Workload Cluster. Latency between Workload Clusters must be below 200ms.

1.3 Preparing the Jumphost

1.3.1 Kubeconfigs

Set up the kubeconfig on the jumphost such that it contains an admin context for each of the three clusters. As these context names may also be incorporated into pod names to distinguish them in the multi-cluster, it is critical to ensure that context names are simple and restricted to lowercase characters and hyphens ('-'). OpenShift generates contexts upon 'oc login' that incorporates the login name, namespace, port numbers, and host name delimited by '/' and ':' characters; do NOT use these contexts when constructing the multi-cluster. In this procedure, we will create contexts named mgmt, workload-a, and workload-b to represent the three clusters. The following best-practice procedure is recommended:

1. Generate a new admin kubeconfig for each of the clusters using the `oc config new-admin-kubeconfig` command. Store each generated kubeconfig in `$HOME/.kube` as `mgmt.yaml`, `workload-a.yaml`, and `workload-b.yaml`, with `0o600` permissions respectively.
2. In each of the kubeconfigs, rename the cluster and context to match the corresponding filenames (sans extension/type) and the user to the same name with a suffix of "-admin" (e.g. `mgmt.-admin`). Dumping each context using the associated kubeconfig should appear as:

```
1 root@jumphost:~# oc --kubeconfig .kube/mgmt.yaml config get-contexts
2 CURRENT NAME CLUSTER AUTHINFO NAMESPACE
3 * mgmt mgmt mgmt-admin
4
5 root@jumphost:~# oc --kubeconfig .kube/workload-a.yaml config get-contexts
6 CURRENT NAME CLUSTER AUTHINFO NAMESPACE
7 * workload-a workload-a workload-a-admin
8
9 root@jumphost:~# oc --kubeconfig .kube/workload-b.yaml config get-contexts
10 CURRENT NAME CLUSTER AUTHINFO NAMESPACE
11 * workload-b workload-b workload-b-admin
```

3. Merge the three kubeconfigs into the default kubeconfig.

```

1 root@jumphost:~# export KUBECONFIG=".kube/mgmt.yaml:.kube/workload-a.yaml:.kube/workload-b.yaml"
2 root@jumphost:~# oc config view --raw --flatten >.kube/config
3 root@jumphost:~# chmod 0o600 .kube/config
4 root@jumphost:~# unset KUBECONFIG ## to use the merged config by default
5 root@jumphost:~# oc config get-contexts
6 CURRENT NAME CLUSTER AUTHINFO NAMESPACE
7 * mgmt mgmt mgmt-admin
8     workload-a workload-a workload-a-admin
9     workload-b workload-b workload-b-admin

```

Utilities

Install the following utilities on the jumphost:

- Helm v3
- Kubernetes CLI (kubectl): 1.25 or later
- Openshift CLI: 4.16 or later

Karmada kubectl plug-in

Install the Karmada kube plug-in with the following commands:

```

1 $ # Download the tarball
2 $ curl -fsSL "https://github.com/karmada-io/karmada/releases/download/v1.13.1/kubectl-karmada-
   linux-amd64.tgz" -o kubectl-karmada-linux-amd64.tgz
3 $ # Extract the contents
4 $ tar xzf kubectl-karmada-linux-amd64.tgz
5 $ # Install the binary
6 $ sudo install kubectl-karmada /usr/local/bin

```

Verify the plug-in version as 1.13.1:

```

1 $ kubectl karmada version
2 kubectl karmada version: version.Info{GitVersion:"v1.13.1", GitCommit:"7
   c4af1bc914f4998893ce3e5b69baf7dc619803b", GitTreeState:"clean", BuildDate:"2025-03-29T03
   :27:17Z", GoVersion:"go1.22.12", Compiler:"gc", Platform:"linux/amd64"}

```

Submariner

Download v0.20.0 of the submariner utility, subctl and install it to /usr/local/bin:

```

1 $ # Download the tarball
2 $ curl -fsSL "https://github.com/submariner-io/releases/releases/download/v0.20.0/subctl-v0
   .20.0-linux-amd64.tar.xz" -o subctl-v0.20.0-linux-amd64.tar.xz
3 $ # Extract the contents
4 $ tar xfJ subctl-v0.20.0-linux-amd64.tar.xz
5 $ # Install the binary
6 $ sudo install subctl-v0.20.0/subctl /usr/local/bin

```

1.4 Preparing the Registries (Air-gapped)

This step is only needed for air-gapped installations.

Pull or transfer the following public images to the jumphost:

```

1 quay.io/submariner/lighthouse-agent:0.20.0
2 quay.io/submariner/lighthouse-coredns:0.20.0
3 quay.io/submariner/nettest:0.20.0
4 quay.io/submariner/submariner-gateway:0.20.0
5 quay.io/submariner/submariner-operator:0.20.0
6 quay.io/submariner/submariner-route-agent:0.20.0
7 registry.k8s.io/kube-apiserver:v1.31.3
8 registry.k8s.io/kube-controller-manager:v1.31.3
9 registry.k8s.io/etcd:3.5.16-0
10 docker.io/nginxinc/nginx-unprivileged:1.27.5

```

```

11 docker.io/karmada/karmada-aggregated-apiserver:v1.13.1
12 docker.io/karmada/karmada-controller-manager:v1.13.1
13 docker.io/karmada/karmada-scheduler:v1.13.1
14 docker.io/karmada/karmada-webhook:v1.13.1
15 docker.io/karmada/karmada-operator:v1.13.1
16 docker.io/karmada/karmada-descheduler:v1.13.1
17 docker.io/karmada/karmada-scheduler-estimator:v1.13.1

```

Create the namespaces that Karmada and Submariner will be deployed to on each cluster:

```

1 oc --context mgmt create ns karmada-system
2 oc --context mgmt create ns submariner-operator
3 oc --context workload-a create ns submariner-operator
4 oc --context workload-b create ns submariner-operator

```

Re-tag the following images for the Management Cluster, taking note that the repository for each image must be adjusted to match the associated namespace name:

```

1 <mgmt.cluster.registry.address>/submariner-operator/submariner-operator:0.20.0
2 <mgmt.cluster.registry.address>/karmada-system/kube-apiserver:v1.31.3
3 <mgmt.cluster.registry.address>/karmada-system/kube-controller-manager:v1.31.3
4 <mgmt.cluster.registry.address>/karmada-system/etcd:3.5.16-0
5 <mgmt.cluster.registry.address>/karmada-system/nginx-unprivileged:1.27.5
6 <mgmt.cluster.registry.address>/karmada-system/karmada-webhook:v1.13.1
7 <mgmt.cluster.registry.address>/karmada-system/karmada-aggregated-apiserver:v1.13.1
8 <mgmt.cluster.registry.address>/karmada-system/karmada-controller-manager:v1.13.1
9 <mgmt.cluster.registry.address>/karmada-system/karmada-scheduler:v1.13.1
10 <mgmt.cluster.registry.address>/karmada-system/karmada-operator:v1.13.1
11 <mgmt.cluster.registry.address>/karmada-system/karmada-descheduler:v1.13.1
12 <mgmt.cluster.registry.address>/karmada-system/karmada-scheduler-estimator:v1.13.1

```

Re-tag the following images for each of the workload clusters, taking note that the repository for each image must be adjusted to match the associated namespace name:

```

1 <workload-a.cluster.registry.address>/submariner-operator/submariner-operator:0.20.0
2 <workload-a.cluster.registry.address>/submariner-operator/lighthouse-agent:0.20.0
3 <workload-a.cluster.registry.address>/submariner-operator/lighthouse-coredns:0.20.0
4 <workload-a.cluster.registry.address>/submariner-operator/nettest:0.20.0
5 <workload-a.cluster.registry.address>/submariner-operator/submariner-gateway:0.20.0
6 <workload-a.cluster.registry.address>/submariner-operator/submariner-operator:0.20.0
7 <workload-a.cluster.registry.address>/submariner-operator/submariner-route-agent:0.20.0
8 <workload-b.cluster.registry.address>/submariner-operator/submariner-operator:0.20.0
9 <workload-b.cluster.registry.address>/submariner-operator/lighthouse-agent:0.20.0
10 <workload-b.cluster.registry.address>/submariner-operator/lighthouse-coredns:0.20.0
11 <workload-b.cluster.registry.address>/submariner-operator/nettest:0.20.0
12 <workload-b.cluster.registry.address>/submariner-operator/submariner-gateway:0.20.0
13 <workload-b.cluster.registry.address>/submariner-operator/submariner-operator:0.20.0
14 <workload-b.cluster.registry.address>/submariner-operator/submariner-route-agent:0.20.0

```

Push all the re-tagged images to the cluster registries.

1.5 Installing Submariner 0.20.0

Submariner is a CNCF Sandbox Project. Submariner enables the interconnection of the internal networks of two K8s clusters through an L3 tunnel. The interconnection enables cluster-internal communication between the application workloads on each cluster.

The Submariner Broker will be installed on the Management Cluster while a Submariner instance will be installed on each of the Workload Clusters. For the Submariner instances to reach the Broker's API, they must be able to resolve the Kubernetes API (e.g. api..) on the Management Cluster. This DNS name needs to be resolved through a proper DNS lookup. Add the following A-records to the DNS server whose IP address was specified during ISO creation:

```

1 api.<mgmtClusterName>.<domain> A <ClusterMgmtIP>
2 *.apps.<mgmtClusterName>.<domain> A <ClusterMgmtIP>

```

From the shell of one of the Workload Cluster nodes (for RHOC P Workload clusters access the shell via ‘oc debug’) perform an nslookup against the DNS name to ensure it resolves.

```
1 sh-5.1# nslookup api.<mgmtClusterName>.<domain>
2 .
3 .
4 Name: api.<mgmtClusterName>.<domain>
5 Address: <ClusterMgmtIP>
```

Next, deploy the broker to the **Management Cluster** context using the management cluster’s kubeconfig:

Air-gapped:

```
1 subctl deploy-broker --context mgmt --repository image-registry.openshift-image-registry.svc
:5000/submariner-operator
```

Non-air-gapped:

```
1 $ subctl deploy-broker ---context mgmt
```

If successful, the broker deployment will create a file called broker-info.subm. Run subctl –context mgmt show brokers to verify the installation.

```
1 [x] Detecting broker(s)
2 NAMESPACE NAME COMPONENTS GLOBALNET GLOBALNET CIDR DEFAULT GLOBALNET SIZE DEFAULT DOMAINS
3 submariner-k8s-broker submariner-broker service-discovery, connectivity no 242.0.0.0/8 65536
```

Now, join each of the WL Clusters together via the Broker:

Air-gapped (add the –repository flag to use the cluster’s private registry):

```
1 $ subctl join --context workload-a --air-gapped --repository image-registry.openshift-image-
registry.svc:5000/submariner-operator --natt=false broker-info.subm
2 $ subctl join --context workload-b --air-gapped --repository image-registry.openshift-image-
registry.svc:5000/submariner-operator --natt=false broker-info.subm
```

Non-air-gapped:

```
1 $ subctl join --context workload-a --air-gapped --natt=false .local/jnpr-multiclust er-submariner
/broker-info.subm
2 $ subctl join --context workload-b --air-gapped --natt=false .local/jnpr-multiclust er-submariner
/broker-info.subm
```

If both WL Clusters are joined successfully, we can verify the Submariner deployment by running a set of unit tests via subctl. These tests take about 5 minutes to run. Disable disruptive verifications when prompted. The following command tests the connectivity from the workload-a Cluster’s K8s context to workload-b Cluster’s context (see kubect l config get-contexts’):

Air-gapped:

```
1 $ # Warnings displayed about "system:authenticated" not being found are expected and can be
ignored
2 $ oc --context workload-a -n submariner-operator policy add-role-to-group system:image-puller
system:authenticated
3 $ oc --context workload-b -n submariner-operator policy add-role-to-group system:image-puller
system:authenticated
```

Verify the installation:

```
1 $ subctl verify --context <workload1Context> --tocontext <workload2Context>
2 ? You have specified disruptive verifications (gateway-failover). Are you sure you want to run
them? (y/N) N
```

Currently, there is a known issue with Submariner tests with RHOC P 4.18 (see [Connectivity test fails with OCP 4.18](#)). We expect the following test failures:

```

1 Summarizing 2 Failures:
2
3 [FAIL] Basic TCP connectivity tests across clusters without discovery when a pod connects via
   TCP to a remote pod when the pod is on a gateway and the remote pod is on a gateway [It]
   should have sent the expected data from the pod to the other pod [dataplane, basic]
4 github.com/submariner-io/shipyard@v0.20.0/test/e2e/tcp/connectivity.go:72
5 [FAIL] Basic TCP connectivity tests across clusters without discovery when a pod connects via
   TCP to a remote service when the pod is on a gateway and the remote service is on a
   gateway [It] should have sent the expected data from the pod to the other pod [dataplane,
   basic]
6 github.com/submariner-io/shipyard@v0.20.0/test/e2e/tcp/connectivity.go:72
7
8 Ran 17 of 48 Specs in 318.692 seconds
9 FAIL! -- 15 Passed | 2 Failed | 0 Pending | 31 Skipped

```

These failures are benign and are to be ignored.

1.6 Installing Karmada 1.13.1

Karmada is a CNCF (Cloud Native Computing Foundation) Incubating Project. Karmada enables workload scheduling across multiple K8s clusters and/or clouds.

The Karmada operator will be used to install Karmada on the Management Cluster. Note that in the yaml definitions below there are certain lines marked as being needed for air-gapped installs, which can be omitted otherwise.

1. Create the 'karmada-system' namespace/project on the Management Cluster

```
1 $ oc new-project karmada-system --context mgmt
```

2. Bind the privileged Security Context Constraint (SCC) to the karmada-operator and default ServiceAccounts (does NOT require the karmada-operator ServiceAccount to exist) in the karmada-system namespace on the management context:

```

1 $ oc --context mgmt -n karmada-system adm policy add-scc-to-user privileged -z karmada-operator
2 $ oc --context mgmt -n karmada-system adm policy add-scc-to-user privileged -z default

```

3. Download the [Karmada v1.13.1 Operator Chart](#) and the CRDs tarballs:

```

1 $ curl -fsSL "https://github.com/karmada-io/karmada/releases/download/v1.13.1/karmada-operator-
   chart-v1.13.1.tgz" -o karmada-operator-chart-v1.13.1.tgz
2 $ curl -fsSL "https://github.com/karmada-io/karmada/releases/download/v1.13.1/crds.tar.gz" -o
   1.13.1-crds.tar.gz

```

4. Create a values file (values.yaml) for deploying the operator:

```

1 global: # For air-gapped
2
3 imageRegistry: "image-registry.openshift-image-registry.svc:5000"
4 installCRDs: false
5 operator:
6 image:
7 repository: karmada-system/karmada-operator # For air-gapped
8 tag: v1.13.1 # Version of container image to pull
9 podAnnotations:
10 openshift.io/required-scc: privileged # Operator needs to store CRs to container FS

```

5. Helm-install the operator with the above values file.

```
1 $ helm --kube-context mgmt install -n karmada-system karmada-operator --values values.yaml ./
   karmada-operator-chart-v1.13.1.tgz --wait
```

6. Verify the operator reaches a running state

```
1 $ kubectl get pods -n karmada-system --context mgmt
2 NAME READY STATUS RESTARTS AGE
3 karmada-operator-6cb799fdcf-792qh 1/1 Running 0 2m
```

7. Once installed, the operator introduces several Custom Resource Definitions (CRDs) that can be used to configure the various Karmada components.

```
1 $ oc explain karmada.spec.components
2 GROUP: operator.karmada.io
3 KIND: Karmada
4 VERSION: v1alpha1
5
6 FIELD: components <Object>
7
8 DESCRIPTION:
9     Components define all of karmada components.
10    not all of these components need to be installed.
11 .
12 .
```

Air-gapped:

- For air-gapped installs, the CRDs tarball must be hosted in an offline webserver accessible by the operator.

```
1 $ # Create a ConfigMap containing the CRD tarball:
2 $ oc --context mgmt -n karmada-system create cm karmada-crds --from-file=./1.13.1-crds.tar.gz
```

- Create a manifest (crd-webserver.yaml) for an Nginx instance serving the CRD tarball via a ClusterIP service inside Karmada's namespace:

```
1 # crd-webserver.yaml
2 apiVersion: v1
3 kind: Service
4 metadata:
5   name: karmada-crds
6   namespace: karmada-system
7 spec:
8   selector:
9     app: karmada-crds
10  type: ClusterIP
11  ports:
12    - name: karmada-crds
13      protocol: TCP
14      port: 80
15      targetPort: 8080
```

- Apply the manifest

```
1 $ oc --context mgmt -n karmada-system apply -f crd-webserver.yaml
```

- Verify that the NGINX instance is in a running state

```
1 $ oc --context mgmt -n karmada-system get pods
2 NAME READY STATUS RESTARTS AGE
3 karmada-crds-667fc8979f-s6jcd 1/1 Running 0 10d
```

8. Create a custom resource YAML, karmada-instance.yaml, file (replacing the certSAN elements for the karmadaAPIServer and the default storage class) as follows:

```
1 apiVersion: operator.karmada.io/v1alpha1
2 kind: Karmada
3 metadata:
4   name: karmada
```

```

5 namespace: karmada-system
6 spec:
7   crdTarball: # Air-gapped
8   httpSource: # Air-gapped
9     url: http://karmada-crds/1.13.1-crds.tar.gz # Air-gapped
10  components:
11    karmadaDescheduler:
12      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
13        descheduler
14      imageTag: v1.13.1
15    karmadaAggregatedAPIServer:
16      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
17        aggregated-apiserver # Air-gapped
18      imageTag: v1.13.1
19      annotations:
20        openshift.io/required-scc: privileged
21      featureGates:
22        Failover: true
23    karmadaControllerManager:
24      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
25        controller-manager # Air-gapped
26      imageTag: v1.13.1
27      featureGates:
28        Failover: true
29    karmadaScheduler:
30      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
31        scheduler # Air-gapped
32      imageTag: v1.13.1
33      featureGates:
34        Failover: true
35    karmadaWebhook:
36      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
37        webhook # Air-gapped
38      imageTag: v1.13.1
39    karmadaMetricsAdapter:
40      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/karmada-
41        metrics-adapter # Air-gapped
42      imageTag: v1.13.1
43      annotations:
44        openshift.io/required-scc: privileged
45    kubeControllerManager:
46      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/kube-
47        controller-manager # Air-gapped
48      imageTag: v1.31.3
49    karmadaAPIServer:
50      imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/kube-
51        apiserver # Air-gapped
52      imageTag: v1.31.3
53      serviceType: NodePort # Expose the API server to the WL clusters with external IP
54      certSANs: # Add SANs so that the WL Clusters can accept the Mgmt Cluster's certificate
55        - <MgmtClusterVIP> # Mgmt Cluster Mgmt VIP/apiserver
56      annotations:
57        openshift.io/required-scc: privileged
58    #
59    # Default storage for etcd will be to use hostPath which
60    # OpenShift will not tolerate so we add a 5Gi PVC
61    # against the default storageClass (longhorn)
62    #
63    etcd:
64      local:
65        imageRepository: image-registry.openshift-image-registry.svc:5000/karmada-system/etcd #
66        Air-gapped
67        imageTag: 3.5.16-0
68        replicas: 3
69        volumeData:
70          volumeClaim:
71            spec:
72              accessModes:
73                - ReadWriteOnce
74              resources:
75                requests:
76                  storage: 5Gi
77              storageClassName: <defaultStorageClass>

```

Note: that for most of the Karmada components, we add an annotation that allows them to run in *privileged* mode. Without these annotations, OpenShift will apply the most restrictive SCC (most interactions with the OS or the container filesystem will be curtailed).

9. Apply the CR

```
1 $ oc --context mgmt apply -n karmada-system -f karmada-instance.yaml
```

10. Monitor the Karmada deployment

```
1 $ oc get pods -n karmada-system -w --context mgmt
2
3 NAME READY STATUS RESTARTS AGE
4 karmada-aggregated-apiserver-7ff75d6979-wgk6z 1/1 Running 0 63m
5 karmada-apiserver-6df7fd8bdc-2dcct 1/1 Running 0 63m
6 karmada-controller-manager-648c44f5c8-pglr9 1/1 Running 0 63m
7 karmada-etcd-0 1/1 Running 0 63m
8 karmada-kube-controller-manager-544f955cbd-79nkd 1/1 Running 0 63m
9 karmada-metrics-adapter-5bb86d4b9-pj6lr 1/1 Running 0 63m
10 karmada-metrics-adapter-5bb86d4b9-qjzgx 1/1 Running 0 63m
11 karmada-scheduler-7bd65745b9-sgkmm 1/1 Running 0 63m
12 karmada-webhook-998985fb-trnt8 1/1 Running 0 63m
13 karmada-operator-6cb799fdcf-792qh 1/1 Running 0 145m
```

11. Once all pods have reached a running state, verify that the health of the Karmada deployment is good

```
1 $ oc get karmada -n karmada-system --context mgmt
2 NAME READY AGE
3 karmada True 86m
```

1.6.1 Preparing the Karmada Context

1. The WL Clusters, as part of the join process, will need to access the Karmada API server. Verify that the API Server has a NodePort address and is reachable:

```
1 $ oc get services -n karmada-system --kubeconfig <mgmtKubeconfig>
2 NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
3 karmada-aggregated-apiserver ClusterIP 172.30.29.244 <none> 443/TCP 91m
4 karmada-apiserver NodePort 172.30.11.192 <none> 5443:31463/TCP 91m
5 karmada-etcd ClusterIP None <none> 2379/TCP,2380/TCP 92m
6 karmada-etcd-client ClusterIP 172.30.62.228 <none> 2379/TCP 92m
7 karmada-metrics-adapter ClusterIP 172.30.37.238 <none> 443/TCP 91m
8 karmada-webhook ClusterIP 172.30.196.143 <none> 443/TCP 91m
9
10 $ nc -zv <mgmtClusterNodeIP> 31463
11 Connection to <mgmtClusterNodeIP> (10.9.177.57) 31463 port [tcp/*] succeeded!
```

2. Fetch the kube config for the newly created Karmada context

```
1 $ oc -n karmada-system get secrets karmada-admin-config -o jsonpath='{.data.karmada\.config}'
   --context mgmt | base64 -d > .kube/karmada.yaml
```

3. Edit the cluster and context names in the karmada kubeconfig to 'karmada-apiserver' and change the cluster address to match the Management Cluster's HA (VIP) address.
4. Add the new kubeconfig to the default kubeconfig (see step 3 in Kubeconfigs for an example of how to flatten the kubeconfigs)

```
1 root@jumphost:~# oc config get-contexts
2 CURRENT NAME CLUSTER AUTHINFO NAMESPACE
3 * karmada-apiserver karmada-apiserver karmada-admin
4   mgmt mgmt mgmt-admin
5   workload-a workload-a workload-a-admin
6   workload-b workload-b workload-b-admin
```

5. The Management Cluster will need to be able to reach each of the WL Clusters through a DNS lookup. Ensure that the DNS Server serving the Management Cluster has entries for the WL Cluster API servers. E.g.

```
1 <WLaClusterMgmtIP> api.<WLaClusterDNSName>
2 <WLbClusterMgmtIP> api.<WLbClusterDNSName>
```

1.6.2 Joining the Workload Clusters

1. Ensure that the Karmada-context is the working context

```
1 $ oc config use-context karmada-apiserver
```

2. Create a certificate-based kubeconfigs for each Workload Cluster:

```
1 $ oc --context workload-a config new-admin-kubeconfig >.kube/wla-admin.yaml
2 $ oc --context workload-b config new-admin-kubeconfig >.kube/wlb-admin.yaml
```

3. JOIN Workload Cluster A - Join the WL Cluster

```
1 $ oc karmada join workload-a --cluster-context workload-a
```

4. Enable the Karmada Scheduler Estimator for Workload Cluster A

```
1 $ oc karmada addons enable karmada-scheduler-estimator --karmada-kubeconfig .kube/config --
  context mgmt -C workload-a --member-kubeconfig .kube/wla-admin.yaml --member-context admin
  --karmada-scheduler-estimator-image='image-registry.openshift-image-registry.svc:5000/
  karmada-system/karmada-scheduler-estimator:v1.13.1'
```

5. JOIN Workload Cluster B - Join the WL Cluster

```
1 $ oc karmada join workload-b --cluster-context workload-b
2   cluster workload-b is joined successfully
```

6. Enable the Karmada Scheduler Estimator for Workload Cluster B

```
1 $ oc karmada addons enable karmada-scheduler-estimator --karmada-kubeconfig .kube/config --
  context mgmt -C workload-b --member-kubeconfig .kube/wlb-admin.yaml --member-context admin
  --karmada-scheduler-estimator-image='image-registry.openshift-image-registry.svc:5000/
  karmada-system/karmada-scheduler-estimator:v1.13.1'
```

7. Verify both constituent clusters of the multi-cluster are READY

```
1 $ kubectl get clusters --context karmada-apiserver
2 NAME VERSION MODE READY AGE
3 workload-a v1.31.6 Push True 123m
4 workload-b v1.31.6 Push True 123m
```

and that the scheduler-estimators are in a running state in the Management Cluster's context, e.g.:

```
1 $ kubectl --context mgmt -n karmada-system get pods
2 NAME READY STATUS RESTARTS AGE
3 karmada-aggregated-apiserver-6ddb95dfb5-vr7kx 1/1 Running 12 (24d ago) 24d
4 karmada-apiserver-6d5498c75c-r6822 1/1 Running 7 (24d ago) 24d
5 karmada-controller-manager-6767545f84-5nnkt 1/1 Running 0 9d
6 karmada-descheduler-56f49d77f5-9dp6w 1/1 Running 1 (24d ago) 24d
7 karmada-etcd-0 1/1 Running 0 24d
8 karmada-etcd-1 1/1 Running 0 24d
9 karmada-etcd-2 1/1 Running 0 24d
10 karmada-kube-controller-manager-797dd75f96-xw6zg 1/1 Running 10 (24d ago) 24d
11 karmada-metrics-adapter-75d478fb7-7964b 1/1 Running 11 (24d ago) 24d
12 karmada-metrics-adapter-75d478fb7-pdqjz 1/1 Running 11 (24d ago) 24d
13 karmada-operator-7c8b9b8f69-wdlhx 1/1 Running 1 (24d ago) 24d
14 karmada-scheduler-55cdf456d9-mxcdz 1/1 Running 0 10d
15 karmada-scheduler-estimator-workload-1-7d595559f4-p5sw9 1/1 Running 0 10d
16 karmada-scheduler-estimator-workload-2-667fc8979f-s6jcd 1/1 Running 0 10d
17 karmada-webhook-6c775f68c8-blprz 1/1 Running 1 (24d ago) 24d
```

1.7 Operational Tips

1.7.1 Application Support

BBE Applications which include support for a multi-geo multi-cluster include:

- BNG CUPS Controller – release 24.4R2 and later
- Address Pool Manager (APM) – release 3.4.0 and later

1.7.2 Karmada kubeconfig

Keep track of the kube config file for the Management Cluster that you generated in step 2 of Preparing the Karmada Context. You will need to use this to generate a secret for the application's Observer micro-service to monitor Karmada scheduling events. See application installation guide for details on creating the kube config secret.

1.7.3 Template File for Multi-geo

Application setup (APM of BNG CUPS Controller) has a lot more values to collect from the operator. At a minimum, registry push/pull addresses and the Karmada kubeconfig secrets can be put in a template file to be passed to the utility script's setup step (`-template`).

In the example below, the Management Cluster's Push FQDN is `default-route-openshift-image-registry.apps.wf-mg-rh-kd-mdr.englab.juniper.net`, Workload-a's Push FQDN is `default-route-openshift-image-registry.apps.wf-mg-rh-wla-mdr.englab.juniper.net` and Workload-b's FQDN is `default-route-openshift-image-registry.apps.wf-mg-rh-wlb-mdr.englab.juniper.net`:

```

1  global:
2    jnpr:
3      karmada:
4        registries:
5          wf-mg-rh-wlb-mdr: image-registry.openshift-image-registry.svc:5000 # backupClusterName:
6            imagePullTransportAddress
7        backup_clusters:
8          - wf-mg-rh-wlb-mdr # - backupClusterName
9        primary_cluster: wf-mg-rh-wla-mdr # primaryClusterName
10       registry:
11         push:
12           - default-route-openshift-image-registry.apps.wf-mg-rh-kd-mdr.englab.juniper.net #
13             MgmtClusterRegistryPushFQDN
14           - default-route-openshift-image-registry.apps.wf-mg-rh-wla-mdr.englab.juniper.net #
15             PrimaryWlClusterRegistryPushFQDN
16           - default-route-openshift-image-registry.apps.wf-mg-rh-wlb-mdr.englab.juniper.net #
17             BackupWlClusterRegistryPushFQDN
18         registry:
19           pull: image-registry.openshift-image-registry.svc:5000 #
20             primaryClusterRegistryPullTransportAddress
21       observer:
22         registry:
23           pull: image-registry.openshift-image-registry.svc:5000 #
24             managementClusterRegistryPullTransportAddress
25       secrets:
26         kubeconfig: karmada-kconf # name of the Karmada kubeconfig secret object

```

1.7.4 Switchover vs Failover

The applications include a 'multi-cluster switchover' command with their utility script. The application's micro-service charts carry an application-specific cluster toleration (1 second). The utility script's switchover

command applies a “NoExecute” taint to the cluster against the application-specific toleration in-order to trigger a switchover event. Micro-services that only exist on one Workload Cluster will be recreated on the other Workload Cluster and de-scheduled on their original Workload Cluster.

Karmada initiates failover procedures when it detects that a workload cluster is no longer viable for running workloads. Micro-service multi-cluster policies are re-evaluated; micro-services that only exist on one Workload Cluster will be re-scheduled on the other Workload Cluster. When the original Workload Cluster becomes reachable, those micro-services will be de-scheduled.

1.7.5 Monitoring and Troubleshooting

Generally running ‘kubectl get clusters’ against the Karmada context will give you an overview of the ready state of the two Workload Clusters it is monitoring. The Ready state of both Workload Clusters should be True.

```
1 $ kubectl get clusters ---context <karmadaContextName>
2 NAME VERSION MODE READY AGE
3 <wlaClusterName> v1.31.6 Push True 64d
4 <wlbClusterName> v1.31.6 Push True 64d
```

Karmada also supports a Prometheus endpoint for access to various alerts and key metrics. The following references are useful for setting up Prometheus monitoring:

- <https://karmada.io/docs/administrator/monitoring/working-with-prometheus-in-control-plane>
- <https://karmada.io/docs/administrator/monitoring/working-with-prometheus>

Application workloads are tracked by ResourceBinding objects in the application namespace of the Karmada context. The ResourceBinding objects provide information on where the workload is scheduled and other useful meta-data about the workload. For example,

```
1 $ kubectl get ResourceBinding -n jnpr-apm ---context <KarmadaContext>
2 NAME SCHEDULED FULLYAPPLIED AGE
3 apm-apmi-<wlaClusterName>-service True True 30h
4 apm-apmi-<wlbClusterName>-service True True 30h
```

lists the resource Bindings for APM. Any ResourceBinding with a FULLYAPPLIED status of False may be suspect and worth delving into by describing the object. For example, describing a ResourceBinding for the provman deployment tells us which Workload Cluster Karmada expects this deployment to be scheduled on.

```
1 $ kubectl describe ResourceBinding -n jnpr-apm jnpr-apm-provman-deployment --context <
   KarmadaContext>
2 Name: jnpr-apm-provman-deployment
3 Namespace: jnpr-apm
4 Labels: propagationpolicy.karmada.io/permanent-id=5c93ec57-e25f-4466-aa54-13f2a6f1289d
5         resourcebinding.karmada.io/permanent-id=fca6e395-b54a-4aff-b5ba-bed8dcbfd6da
6 Annotations: policy.karmada.io/applied-placement:
7               {"clusterAffinities":[{"affinityName":"primary-clusters","clusterNames":["<workload-
8               a>"]}, {"affinityName":"backup-clusters","clusterNam...
9               propagationpolicy.karmada.io/name: provman-prop
10              propagationpolicy.karmada.io/namespace: jnpr-apm
11              resourcebinding.karmada.io/dependencies: null
12 API Version: work.karmada.io/v1alpha2
13 Kind: ResourceBinding
14 Status:
15   Aggregated Status:
16     Applied: true
17     Cluster Name: <workloadClusterName>
18     Health: Healthy
19     Status:
20       Available Replicas: 1
21       Generation: 1
```

```

22     Observed Generation: 1
23     Ready Replicas: 1
24     Replicas: 1
25     Resource Template Generation: 2
26     Updated Replicas: 1
27     Conditions:
28       Last Transition Time: 2025-06-05T14:38:39Z
29       Message: All works have been successfully applied
30       Reason: FullyAppliedSuccess
31       Status: True
32       Type: FullyApplied
33       Last Transition Time: 2025-06-05T14:38:39Z
34       Message: Binding has been scheduled successfully.
35       Reason: Success
36       Status: True
37       Type: Scheduled
38     Last Scheduled Time: 2025-06-05T14:38:39Z
39     Scheduler Observed Generation: 3
40     Scheduler Observing Affinity Name: primary-clusters

```

1.8 Conclusion

The construction of a multi-geography multi-cluster from three separate single-geography Kubernetes clusters enables control-plane redundancy for Broadband Edge applications such as BNG CUPS Controller and Address Pool Manager. In the multi-geography multi-cluster, the clusters take on one of two roles: Karmada or Management Cluster, and Workload Cluster. The two Workload Clusters run the bulk of the application workloads. The cluster-internal networks of the two Workload Clusters are interconnected by a layer-3 secure tunnel established by Submariner to enable pod-to-pod communications. The Management Cluster monitors the state of the Workload clusters and the workloads (applications) that are running on them. Should one workload cluster become unviable for supporting workloads consistent with the propagation policies defined in their Helm charts, Karmada will try to honor the propagation policy on the other workload cluster (failover).

1.9 Useful links

- Karmada - <https://karmada.io/>
- Submariner - <https://submariner.io/>
- CNCF Projects - <https://www.cncf.io/projects/>
- BNG CUPS Controller Installation Guide – <https://www.juniper.net/documentation/us/en/software/bng-cups/cups-controller-installation-guide/index.html>
- APM Installation Guide - <https://www.juniper.net/documentation/us/en/software/apm/apm-install-upgrade/topics/topic-map/apm-install-upgrade.html>

1.10 Glossary

- APM – Address Pool Manager
- BBE – Broadband Edge
- BNG – Broadband Network Gateway
- CIDR – Classless Inter-domain routing
- CNF – Cloud-native Network Function
- CNCF – Cloud Native Computing Foundation
- CNI – Container Network Interface

- CR – Custom Resource
- CRD – Custom Resource Definition
- CSI – Container Storage Interface
- CUPS – Control/User Plane Separation
- FQDN – Fully Qualified Domain Name
- K8s – Kubernetes
- NLB – Network Load Balancer
- OCP – Openshift Container Platform (see RHOC)
- OSS – Open-Source Software
- REST – Representational State Transfer
- RHOC – Red Hat Openshift Container Platform
- SCC – Security Context Constraint
- WL – Workload
- YAML - data serialization language (YAML Ain't Markup Language)
- VIP – Virtual IP

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